

FILTER DESIGN TEST CASE

Ladder Filter - WiFi 2.4 GHz for 5G NR Sub-6 GHz Front-End Module
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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Priority:	P2 - High

1. OBJECTIVE

This document describes the design, simulation, and characterization of a Ladder Filter filter targeting WiFi 2.4 GHz for 5G NR Sub-6 GHz Front-End Module. The filter is designed on PZT on Si substrate with a target center frequency of 5297.7 MHz and bandwidth of 337.1 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the 5G NR Sub-6 GHz Front-End Module module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Ladder Filter
- Frequency Band: WiFi 2.4 GHz
- Application: 5G NR Sub-6 GHz Front-End Module
- Substrate: PZT on Si
- Package: CSP (Chip Scale Package)
- Design Tool: Cadence AWR Microwave Office
- Design Center: Newbury Park Design Center

This specification applies to all variants of the Ladder Filter design targeting WiFi 2.4 GHz. Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	5297.7 MHz
Bandwidth (3 dB):	337.1 MHz
Insertion Loss Target:	≤ 3.5 dB
Return Loss Target:	≥ 11.0 dB
Out-of-Band Rejection:	≥ 47.0 dB
Isolation (Duplexer):	≥ 40.0 dB
Q Factor:	5434
Temperature Coefficient:	-15.7 ppm/°C
Die Size:	1.7 x 0.9 mm
Wafer Size:	6-inch
Number of Resonators:	10
Electrode Material:	Pt/Al

Passivation: SiO2

4. TEST PROCEDURE

1. Open Cadence AWR Microwave Office and load the Ladder Filter template project.
2. Define the substrate stack: PZT on Si with specified layer thicknesses.
3. Set design targets: center freq 5297.7 MHz, BW 337.1 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 3.5 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (CSP (Chip Scale Package)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, PZT on Si).
10. Perform on-wafer probing using Rohde & Schwarz FSW Signal Analyzer.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 15893 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 3.5 dB
- Return loss in passband shall be >= 11.0 dB
- Out-of-band rejection at +/- 506 MHz offset >= 47.0 dB
- Group delay variation across passband shall be <= 2 ns
- Temperature drift over -40°C to +85°C shall be <= 10.4 MHz
- Power handling shall be >= +28 dBm at 1 dB compression
- ESD tolerance >= 2000 V (HBM)
- Die yield on qualification lot >= 92%

6. TEST RESULTS

Measurement Date: 2024-09-14
Devices Tested: 27
Insertion Loss (meas): 3.25 dB
Return Loss (meas): 13.4 dB
Rejection (meas): 49.8 dB
Isolation (meas): 44.7 dB
Q Factor (meas): 5434
Group Delay Variation: 6.3 ns
Power Handling: +33 dBm
Die Yield: 94.9%

Result Classification: NEEDS REVIEW

7. OBSERVATIONS AND FINDINGS

- a) The Ladder Filter design demonstrated below-target performance for WiFi 2.4 GHz.
- b) Insertion loss is marginally above target; electrode thickness optimization recommended.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Temperature stability was within specification across full range.
- e) Batch-to-batch reproducibility confirmed over 3 wafer lots.

8. CORRECTIVE ACTIONS

- 1. Review near-band rejection margin for worst-case temperature.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

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